

Shi Yu

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Education

IIIS, Tsinghua University

Aug 2025 – Present

Undergraduate Study

IIIS, Tsinghua University

Feb 2024 – Jun 2025

Pre-College Program

Research Interests

LLMs(LLM Agent), Reinforcement Learning

Research Experience

Undergraduate Research in LLM Agent

Beijing, China June

Supervised by Professor Yi Wu, Yu Wang, Chao Yu, Tsinghua University

2025 – Present

- Evaluated the performance of Large Language Models (LLMs) on International Mathematical Olympiad (IMO) problems and developed a multi-step reasoning system for enhanced solving capabilities.
- Measured the core competencies of LLMs on advanced reasoning tasks including IMO and IPHO questions by employing an ICPL (In-context Preference Learning) framework, which incorporated human feedback.
- Participated in the WideSeek-R1 project(Use wide scaling method to get better performance on WideSearch Dataset) by constructing large-scale training data and doing experiments in RL field for broad information-seeking tasks.
- Investigated the CL-bench benchmark and explored context management strategies to improve LLM performance on continual learning and real-world long-context reasoning tasks.

Undergraduate Research in RLHF

Beijing, China Feb

Supervised by Professor Yi Wu and Yu Wang, Tsinghua University

2025 – Jun 2025

- Gained hands-on research experience in RLHF by investigating and enhancing the reinforcement learning efficacy of LLMs trained on non-transitive human preference datasets.

Robotics Intern

Shenzhen, China Aug

Supervised by Yuchuan Wang, Tencent

2025

- Investigated robotics applications by leveraging a large language model (LLM) to successfully complete a robot indoor navigation task.

Publications

Extending Test-Time Scaling: A 3D Perspective with Context, Batch, and Turn 2025

Chao Yu, Qixin Tan, Jiaxuan Gao, Shi Yu, Hong Lu, Xinting Yang, Zelai Xu, Yu Wang, Yi Wu, Eugene Vinitzky

[arXiv:2511.15738](#) 

WideSeek-R1: Exploring Width Scaling for Broad Information Seeking via Multi-Agent Reinforcement Learning 2026

Zelai Xu, Zhexuan Xu, Ruize Zhang, Chunyang Zhu, Shi Yu, Weilin Liu, Quanlu Zhang, Wenbo Ding, Chao Yu, Yu Wang

[arXiv:2602.04634](#) 

Awards

Freshman Scholarship (Second Prize) 2025
Tsinghua University

Gold Medal (Top 10) 2023
40th Chinese Physics Olympiad

Silver Medal 2022
39th Chinese Physics Olympiad

Technical Skills

Programming Languages: Python, C/C++, Bash

Tools & Frameworks: PyTorch, Git, Github, LaTeX, Docker

Languages: English (Understand), Chinese (Native)